

```

// Error! No implicit conversion
let integer: u8 = decimal;
// FIXME ^ Comment out this line

// Explicit conversion
let integer = decimal as u8;
let character = integer as char;

// Error! There are limitations in conversion rules.
// A float cannot be directly converted to a char.
let character = decimal as char;
// FIXME ^ Comment out this line

println!("Casting: {} -> {} -> {}", decimal, integer,
character);

// when casting any value to an unsigned type, T,
// T::MAX + 1 is added or subtracted until the value
// fits into the new type

// 1000 already fits in a u16
println!("1000 as a u16 is: {}", 1000 as u16);

// 1000 - 256 - 256 - 256 = 232
// Under the hood, the first 8 least significant bits
(LSB) are kept,
// while the rest towards the most significant bit (MSB)
get truncated.
println!("1000 as a u8 is : {}", 1000 as u8);
// -1 + 256 = 255
println!(" -1 as a u8 is : {}", (-1i8) as u8);

// For positive numbers, this is the same as the
modulus
println!("1000 mod 256 is : {}", 1000 % 256);

// When casting to a signed type, the (bitwise) result
is the same as
// first casting to the corresponding unsigned type. If
the most significant

```